JOHARI MBBS ANATOMY



VOLUME I

ANATOMY CHAMPIONS 2.0

MBBS मतलब JOHARI MBBS

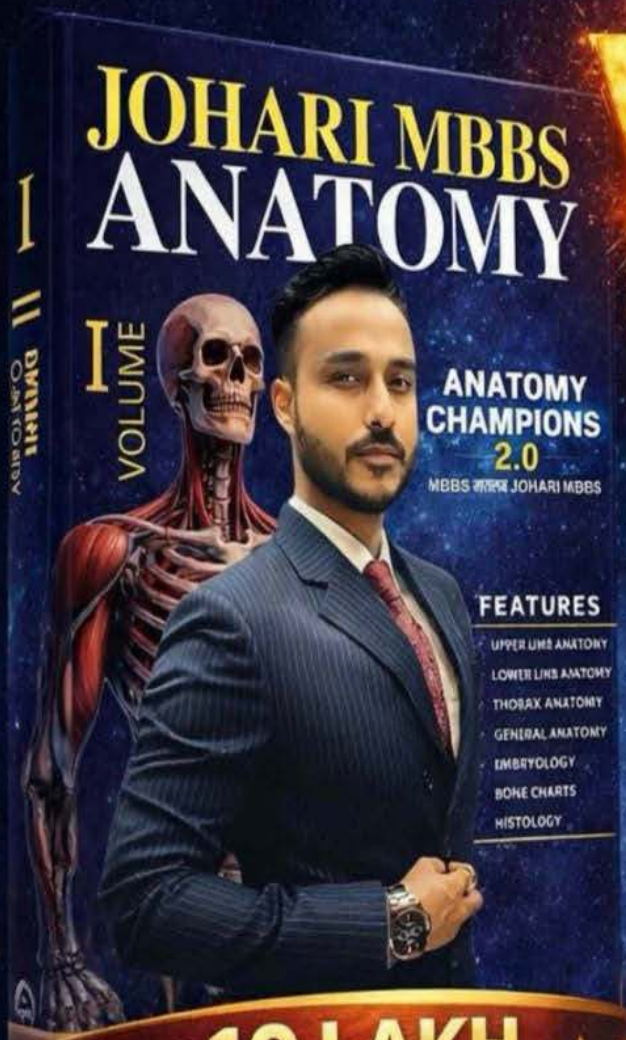
FEATURES

- ✓ UPPER LIMB ANATOMY
- ✓ LOWER LIMB ANATOMY
- ✓ THORAX ANATOMY
- ✓ GENERAL ANATOMY
- ✓ EMBRYOLOGY
- ✓ BONE CHARTS
- ✓ HISTOLOGY

TRUSTED BY

1 MILLION MEDICOS

NEW VERSION



+10 LAKH
STUDENTS TRUSTED

10 LAKH
STUDENTS TRUSTED

VS

OLD VERSION



+2 LAKH
STUDENTS TRUSTED

+2 LAKH
STUDENTS TRUSTED

OLD VERSION

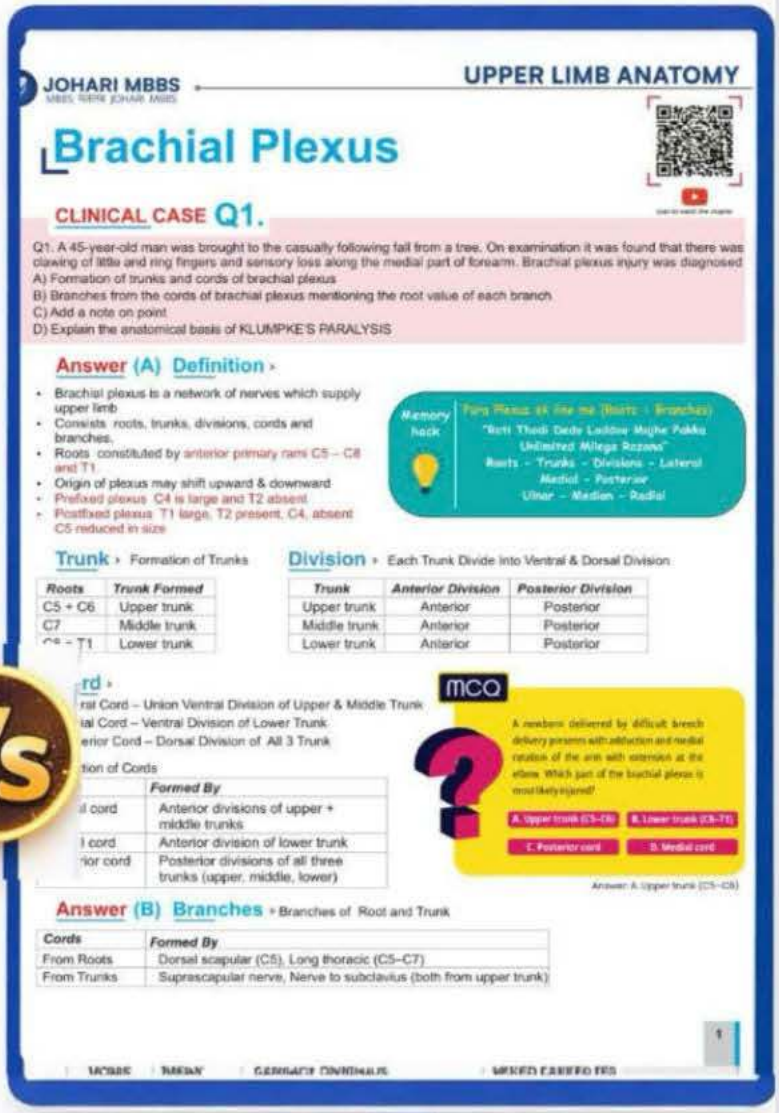
❌ No MCQ Box

VS

NEW VERSION ✅

✅ MCQ Integrated





VS

❌ No MCQ Box

❌ No Clinical Integration

❌ Plain Theory

➡

✅ MCQ Integrated

✅ Clinical Case Added

✅ Exam-Oriented Question Format

CALL : 9827765060 For More Information

GENERAL ANATOMY

Epiphysis

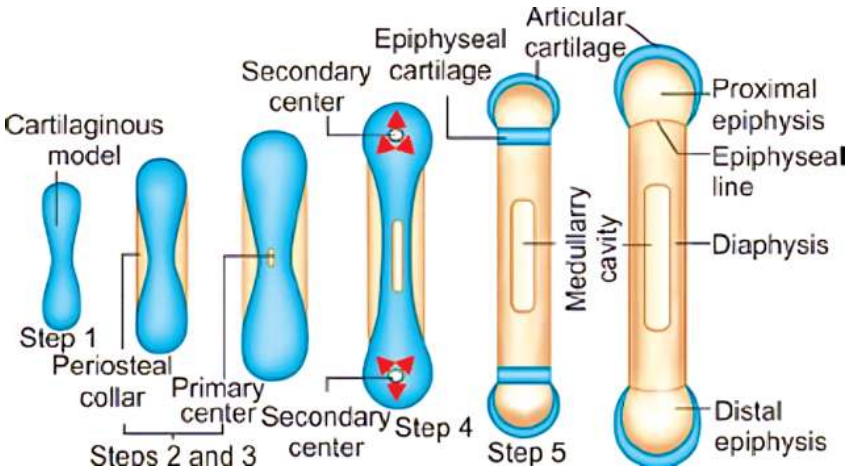
Q. Define Epiphysis along with its types and clinical importance.

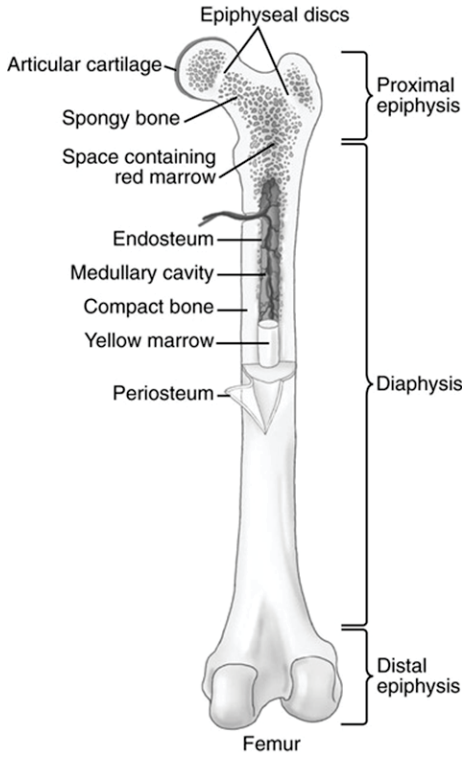
Definition

- The epiphysis is the end part of a long bone, which ossifies from a secondary center of ossification.
- t is separated from the shaft (diaphysis) by a cartilaginous growth plate (epiphyseal plate) during growth, and later fuses with the shaft at maturity.

Characteristics of Epiphysis

- Always located at the ends of long bones.
- Covered with articular cartilage if it forms part of a joint.
- Ossifies later than the diaphysis (from secondary ossification center).
- Plays a key role in longitudinal growth of bone until fusion occurs.
- Fuses with diaphysis at a predictable age





Types of Epiphysis

1. ***Based on Number of Ossification Centers***

- Simple Epiphysis – develops from a single ossification center. Example: Lower end of femur.
- Compound Epiphysis – develops from two or more centers which unite to form one epiphysis. Example: Upper end of humerus.

2. ***Based on Function***

- ***Pressure Epiphysis –***

Articular, transmits weight. Example: Head of femur.

- ***Traction Epiphysis –***
  Non-articular, attachment to tendons. Example: Trochanters of femur.
- ***Atavistic Epiphysis –***
  Evolutionary remnant. Example: Coracoid process of scapula.
- ***Aberrant Epiphysis –***
  Not always present. Example: Head of 1st metacarpal.
- ***Compound Epiphysis (Functional) –***
  Multiple small epiphyses unite before fusing. It is not a true epiphysis and constant
  Example: Upper end of humerus.

Differences Between Epiphysis and Diaphysis

Feature	*Epiphysis*	*Diaphysis*
Location	Ends of long bone	Shaft of long bone
Ossification	Secondary center	Primary center
Function	Growth in length, articulation	Provides strength & support
Bone structure	Mostly cancellous bone inside	Thick cortical compact bone
Fusion	Fuses later with diaphysis	Forms early and remains shaft

Clinical Importance of Epiphysis

i. Growth Indicator – Epiphyseal plate is responsible for longitudinal growth.

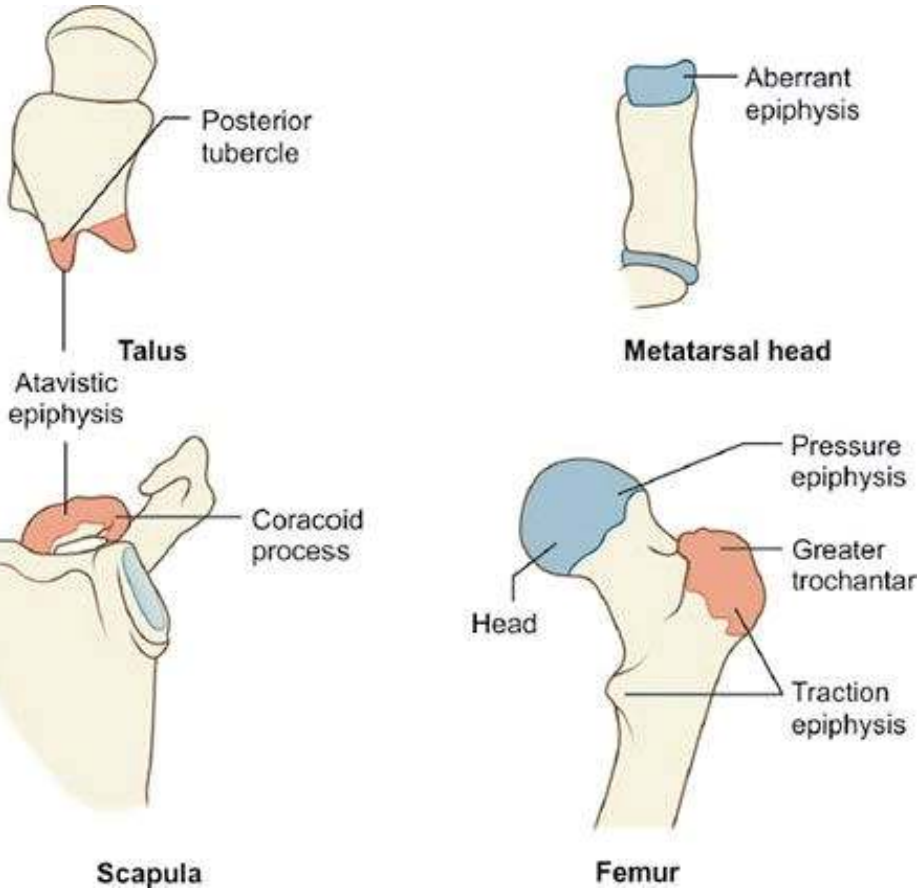
ii. Age Determination – Time of appearance and fusion used for medico-legal age estimation.

iii. Fractures – Epiphyseal injuries (Salter–Harris) are common in children.

iv. Diseases – Rickets, scurvy, achondroplasia affect epiphyseal region.

Summary Table of Types with Examples

Type of Epiphysis	*Examples*
Simple	Lower end of femur, upper end of tibia
Compound	Upper end of humerus
Pressure	Head of femur, head of humerus
Traction	Greater trochanter of femur, tubercles of humerus
Atavistic	Coracoid process of scapula
Aberrant	Head of 1st metacarpal, base of 5th metatarsal



Bone

Q. Explain composition structure and classification of bone

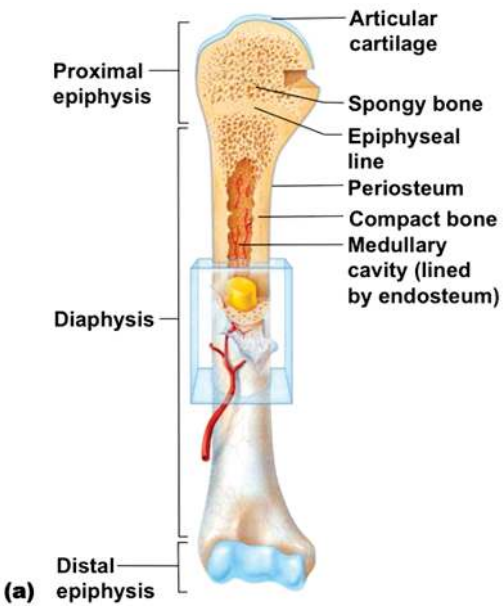
Definition

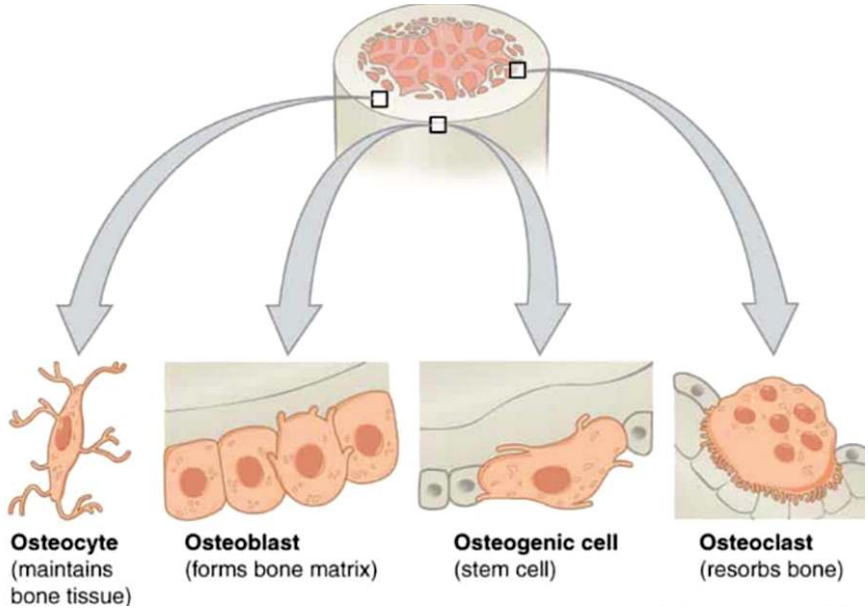
- Bone is a highly specialized connective tissue that forms the main component of the skeletal system.
- It is rigid due to calcified extracellular matrix, but dynamic as it undergoes constant remodeling, resorption, and deposition throughout life.
- Together with cartilage, it forms the skeleton which supports, protects, and facilitates movement.

Composition of Bone

1. ***Organic Component (~30%)***
   - Collagen fibers (mainly type I collagen): Provide tensile strength.
   - Ground substance: Proteoglycans, glycoproteins, osteonectin, osteocalcin → provide elasticity.
   - Cells:
     - Osteoblasts → bone-forming.
     - Osteocytes → mature bone cells, maintain bone matrix metabolism.
     - Osteoclasts → multinucleated giant cells responsible for bone resorption.

2. ***Inorganic Component (~70%)***
   - Mainly calcium and phosphate in the form of hydroxyapatite crystals ($Ca_{10}(PO_4)_6(OH)_2$).
   - Provides hardness and resistance to compression.
   - Also contains sodium, magnesium, carbonate, citrate, fluoride.



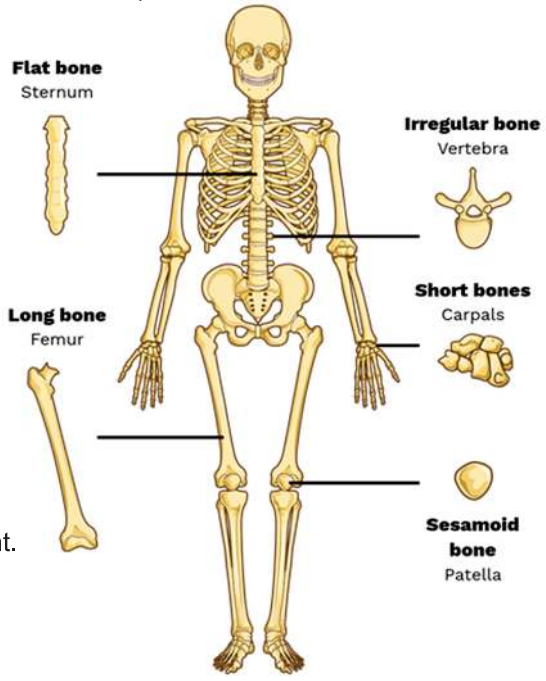


Structure of Bone

1. ***Macroscopic***
   - Compact (Cortical) Bone → Dense, ivory-like, forms outer shell, provides strength.
   - Cancellous (Spongy/Trabecular) Bone → Meshwork of trabeculae filled with red/yellow marrow, lighter and hematopoietic.
2. ***Microscopic***
   - Lamellar Bone (Mature)
     - Organized in concentric lamellae around Haversian canals (osteons).
     - Strong and durable.
   - Woven Bone (Immature)
     - Randomly arranged collagen fibers, less strong.
     - Seen in fetal life, fracture healing, and pathological states (osteosarcoma).

Classification of Bones

A. According to Shape



1. ***Long Bones***
   - Shaft (diaphysis) + two expanded ends (epiphyses).
   - Provide leverage and support weight.
   - Examples: Femur, humerus, tibia.
2. ***Short Bones***
   - Cube-shaped, spongy bone core with thin cortex.
   - Provide stability with limited movement.
   - Examples: Carpals, tarsals.
3. ***Flat Bones***
   - Two compact layers with dipole in between.
   - Protect organs, provide large surface area for muscle attachment.
   - Examples: Skull, sternum, ribs.
4. ***Irregular Bones***
   - Complex shapes, often protective and supportive.
   - Examples: Vertebrae, hip bone, sphenoid.
5. ***Sesamoid Bones***
   - Develop in tendons, modify force of pull, reduce friction.
   - Examples: Patella, pisiform.
6. ***Pneumatic Bones***
   - Contain air-filled cavities lined by mucous membrane.
   - Make bone light and act as resonating chambers.
   - Examples: Maxilla, sphenoid, frontal, ethmoid.
7. ***Accessory Bones***
   - Extra bones formed from ununited ossification centers.
   - Examples: Accessory navicular, sutural (wormian) bones.

B. According to Development

1. ***Membrane (Dermal) Bones → Intramembranous ossification.***
   - Examples: Flat skull bones, clavicle.

2. ***Cartilaginous Bones → Endochondral ossification.***
   - Examples: Long bones like femur, humerus.
3. ***Membrano-cartilaginous Bones → Partly in membrane, partly in cartilage.***
   - Example: Clavicle.
4. ***Somatic Bones → Derived from mesoderm, majority of skeleton.***
5. ***Visceral Bones → Derived from pharyngeal arches.***
   - Examples: Ear ossicles, hyoid bone, styloid process.

C. According to Region

1. ***Axial Skeleton***
   - Skull, vertebrae, ribs, sternum.
   - Provides central support and protects vital organs.
2. ***Appendicular Skeleton***
   - Pectoral girdle, pelvic girdle, upper and lower limb bones.
   - Provides mobility and manipulation.

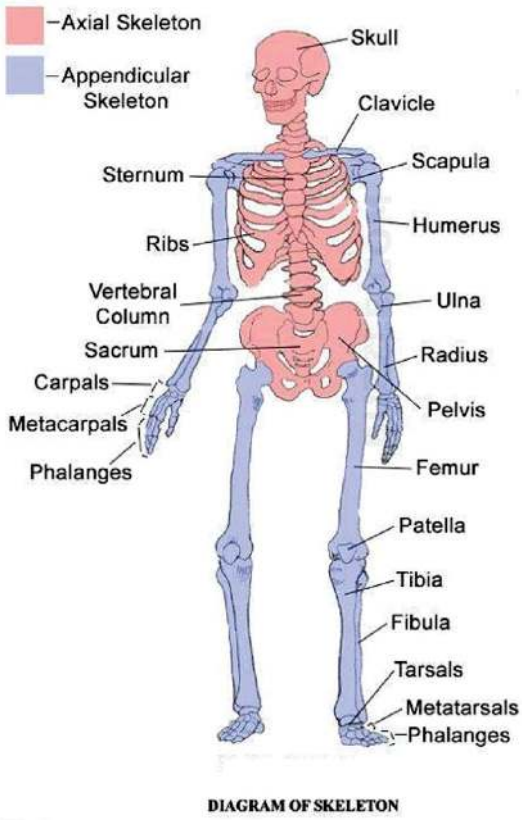
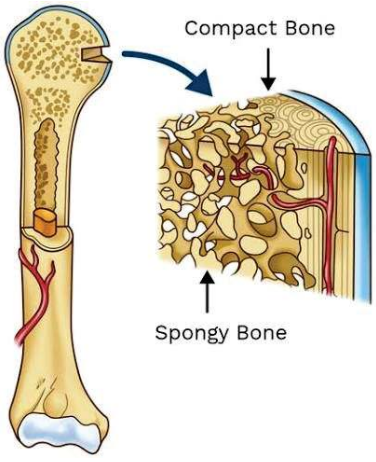
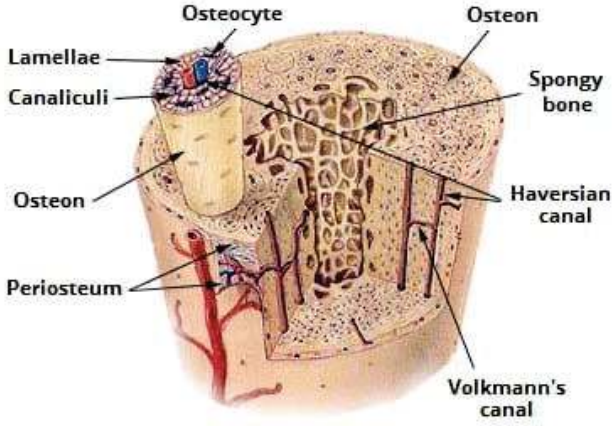


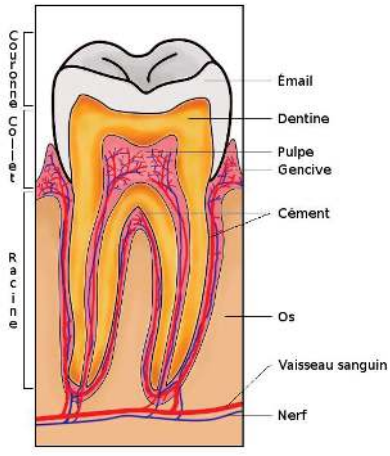
DIAGRAM OF SKELETON

D. According to Microscopic Structure

1. ***Compact Bone***
   - Dense, organized in osteons.
   - Provides mechanical strength.
2. ***Cancellous (Spongy) Bone***
   - Trabecular network, contains bone marrow.
   - Provides hematopoiesis and shock absorption.
3. ***Lamellar Bone***
   - Mature organized bone.
4. ***Woven Bone***
   - Immature, unorganized, temporary (replaced later by lamellar).
5. ***Fibrous Bone***
   - Primitive, seen in early fetal life.
6. ***Dentine & Cement***
   - Tooth-supporting hard tissues, related structurally to bone.







Ossification of Bone

- Two major processes:
  - i. Intramembranous Ossification → Direct from mesenchyme (flat bones, clavicle).
  - ii. Endochondral Ossification → From cartilage template (long bones).
- Primary Ossification Center → Appears early in diaphysis.
- Secondary Ossification Centers → Appear later in epiphyses.
- Fusion of epiphysis with diaphysis marks end of longitudinal growth (puberty-adolescence).

Functions of Bone

- Support → Provides rigid framework.
- Protection → Skull (brain), ribs (lungs/heart), vertebrae (spinal cord).
- Movement → Lever system with muscles.
- Mineral Storage → Reservoir for Ca, P, Mg.
- Hematopoiesis → Red bone marrow produces blood cells.
- Shape & Posture → Maintains body configuration.
- Sound Resonance → Pneumatic bones act as resonators.

Clinical Importance

1. ***Age Estimation***
   - Appearance and fusion of epiphysis → medico-legal use.
2. ***Bone Diseases***
   - Rickets/Osteomalacia → defective mineralization.
   - Osteoporosis → decreased bone mass.
   - Paget's disease → abnormal bone remodeling.
3. ***Fracture Healing***
   - Involves hematoma → soft callus → hard callus → remodeling.
4. ***Bone Marrow Disorders***
   - Anemia, leukemia diagnosed from marrow aspiration/biopsy.
5. ***Pathological Conditions***
   - Osteomyelitis (bone infection), achondroplasia (growth disorder).

Blood Supply of Bones

Q. Write the Blood Supply of Bones

General Principles

- Bone is a living tissue, highly vascular.
- Receives blood supply from multiple sources: nutrient, periosteal, metaphyseal, and epiphyseal arteries.
- Adequate blood supply is essential for growth, remodeling, repair, and vitality of bone.
- Disturbance of blood supply → bone necrosis (osteonecrosis).

Arterial Supply of Long Bones

1. Nutrient Artery

- The chief source of blood supply for the bone.
- Enters through the nutrient foramen (directed away from the growing end → "towards elbow, away from knee").
- Divides into ascending and descending branches within the medullary cavity.
- Forms hairpin loops at the metaphysis and epiphysis.

Supplies:

- Inner 2/3rd of cortex.
- Medullary cavity (bone marrow).
- Metaphysis.

Clinical Note:

- Nutrient artery obstruction → avascular necrosis (common in femoral head, scaphoid).
- Osteomyelitis can occur when septic emboli block nutrient artery.

2. Periosteal Arteries

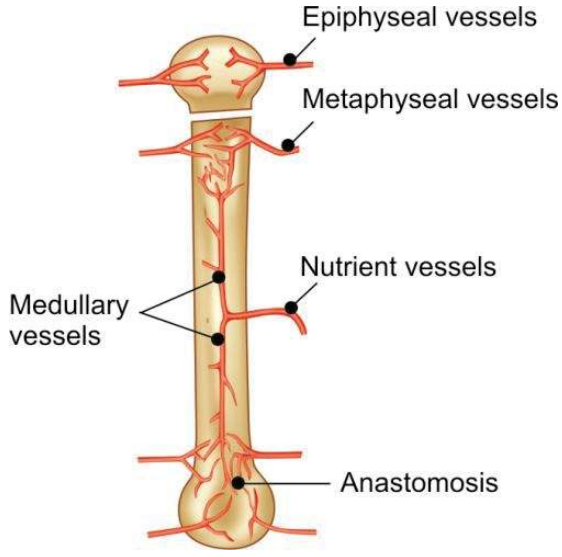
- Numerous small vessels beneath muscular and ligamentous attachments.
- Enter through Volkmann's canals.
- Supply outer 1/3rd of cortex.

3. Epiphyseal Arteries

- Derived from periarticular vascular arcades around joints.
- Supply the epiphysis, especially articular ends.
- Important in the nutrition of growing ends in children.

4. Metaphyseal Arteries

- Arise from nearby systemic vessels (periarticular anastomosis).
- Enter the metaphysis, reinforcing nutrient artery branches.
- Important for metaphyseal circulation and growth plate vascularity.



Blood Supply of Different Bones

A. Long Bones

- Supplied mainly by nutrient artery + periosteal arteries.
- Major share:
  - i. Inner part → nutrient artery.
  - ii. Outer part → periosteal arteries.

B. Short Bones

- Supplied by numerous periosteal arteries that enter non-articular surfaces.
- Since no medullary cavity, no dominant nutrient artery.

C. Vertebrae

- Body: Supplied by anterior & posterior central vessels.
- Arch & processes: Supplied by segmental vessels entering near the base of transverse processes.

D. Ribs

- Supplied by:
  - o Nutrient artery (enters near tubercle).
  - o Periosteal arteries.

Venous Drainage of Bone

- Large, thin-walled veins present in marrow cavity → drain into systemic veins.
- Cortical veins: Accompany arteries in Volkmann's canals.
- Venous drainage is abundant → reason bone is common site for pyogenic infection & metastasis (Batson's plexus in vertebrae).

Lymphatic Drainage

- Bones have sparse lymphatics in periosteum.
- Marrow cavity generally lacks lymphatics.

Nerve Supply

- Supplied by vasomotor nerves accompanying arteries.
- Periosteum richly innervated → main source of pain in fractures, infections, tumors.

Clinical Importance

1. ***Fracture Healing***
   - Rich blood supply → faster healing.
   - Poor vascularity (scaphoid, femoral head, talus) → delayed union, non-union.
2. ***Osteomyelitis***
   - Infection spreads rapidly due to sluggish metaphyseal circulation in children.
3. ***Avascular Necrosis (AVN)***
   - Interruption of epiphyseal or metaphyseal arteries → necrosis (hip, scaphoid, talus).
4. ***Bone Marrow Metastasis***
   - Venous channels allow spread of malignant cells (esp. vertebrae via Batson's plexus).

Student Key Note Space

Cartilage

Q. Explain Classification of Cartilage

Definition

- Cartilage is a specialized connective tissue composed of cells (chondrocytes), fibers, and firm gel-like extracellular matrix rich in proteoglycans and glycosaminoglycans.
- It is avascular, receives nutrition by diffusion from surrounding perichondrium or synovial fluid.
- Intermediate in rigidity between bone and dense connective tissue.

General Features

- Cells: Chondrocytes (mature cells) and chondroblasts (immature cells).
- Matrix: Rich in proteoglycans (aggrecan, chondroitin sulfate), collagen, elastin fibers.
- Covering: Most cartilage (except articular cartilage & fibrocartilage) is surrounded by perichondrium.
- Nutrition: By diffusion through matrix (slow healing).
- Growth:
  - Appositional growth → from perichondrium.
  - Interstitial growth → division of chondrocytes within matrix.

Classification of Cartilage

1. Hyaline Cartilage

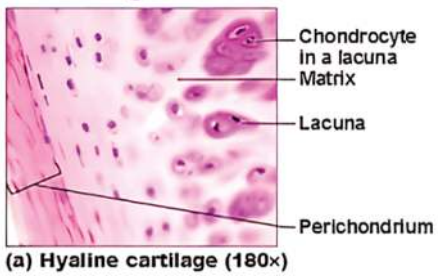
- Location: Articular cartilage of long bones, epiphyseal plates, nasal cartilage, thyroid cartilage, trachea, bronchi, costal cartilage.
- Appearance: Bluish-white, shiny or translucent.
- Fibers: Very thin type II collagen fibers (not visible under light microscope).
- Properties: Smooth, flexible, provides low-friction surface for joints.
- Perichondrium: Present (except at articular surfaces).

2. Fibrocartilage

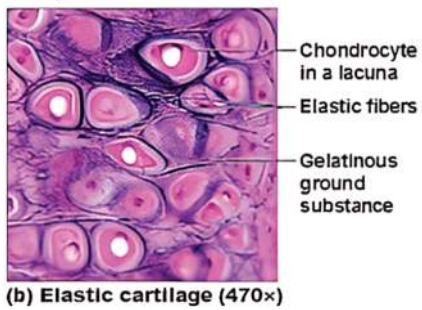
- Location: Intervertebral discs, pubic symphysis, temporomandibular joint, sternoclavicular joint, menisci.
- Appearance: Glistening white, opaque.
- Fibers: Numerous thick type I collagen fibers arranged in bundles.
- Properties: Strongest cartilage, resists compression and shear.
- Perichondrium: Absent.

3. Elastic Cartilage

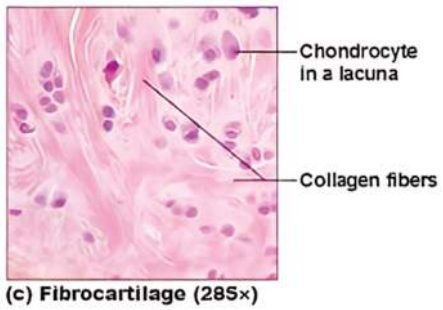
- Location: External ear (pinna), external auditory meatus, eustachian tube, epiglottis.
- Appearance: Yellowish, opaque.
- Fibers: Abundant elastic fibers with type II collagen.
- Properties: Most flexible, maintains shape while allowing bending.
- Perichondrium: Present.



(a) Hyaline cartilage (180×)



(b) Elastic cartilage (470×)



(c) Fibrocartilage (285×)

Differences Between Types of Cartilage

Feature	*Hyaline*	*Fibrocartilage*	*Elastic Cartilage*
Color/Appearance	Bluish-white, shiny	Glistening white, opaque	Yellowish, opaque
Fibers	Thin type II collagen	Thick type I collagen	Elastic + type II collagen
Flexibility	Flexible, smooth	Very strong, less flexible	Most flexible
Perichondrium	Present (except articular)	Absent	Present
Example	Articular cartilage, trachea	Intervertebral disc, pubic symphysis	Pinna, epiglottis

Functions of Cartilage

1. Support: Maintains shape of flexible structures (ear, nose, trachea).
2. Smooth Articulation: Hyaline cartilage reduces friction in joints.
3. Growth of Long Bones: Epiphyseal plate (hyaline cartilage) responsible for longitudinal growth.
4. Shock Absorption: Fibrocartilage (intervertebral discs, pubic symphysis).
5. Elasticity: Elastic cartilage allows bending and recoil (pinna, epiglottis).
6. Template for Ossification: Many bones develop from cartilage model (endochondral ossification).

Clinical Importance

- Limited Healing: Cartilage is avascular, so injuries heal slowly.
- Osteoarthritis: Degeneration of articular cartilage.
- Achondroplasia: Genetic disorder affecting growth plate cartilage → dwarfism.
- Cauliflower Ear: Repeated trauma damages auricular elastic cartilage.
- Intervertebral Disc Herniation: Fibrocartilage degeneration → slipped disc.

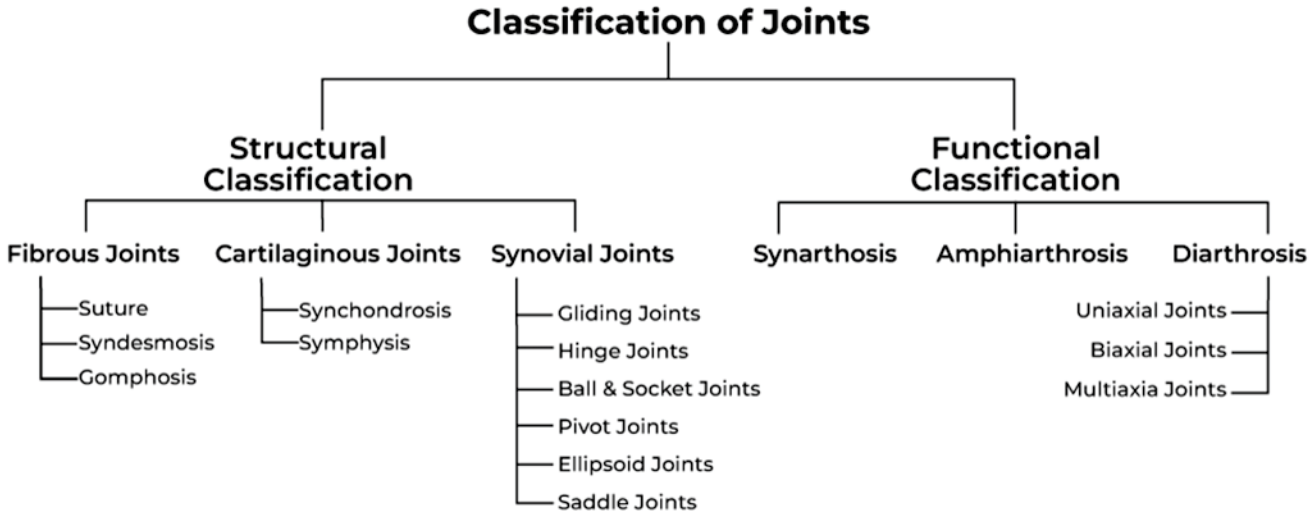
Student Key Note Space

Joints

Q. Write the Short Note on Classification of Joints.

Definition

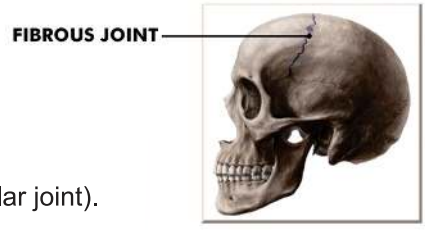
- A joint (articulation) is a junction between two or more bones or cartilages.
- Its function is to permit movement or provide stability.
- Some joints are immovable (skull), while others are highly mobile (shoulder).



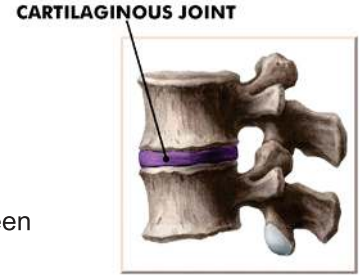
Classification of Joints

A. Structural Classification

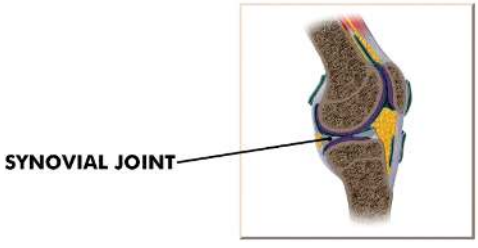
1. ***Fibrous Joints (immovable or slightly movable)***
   - Bones are united by fibrous tissue.
   - Types:
     i. Sutures: Found only in skull, immovable.
     ii. Syndesmosis: Fibrous union between long bones (e.g., inferior tibiofibular joint).
     iii. Gomphosis: Peg-and-socket joint between tooth and socket of jaw.



2. ***Cartilaginous Joints (slightly movable)***
   - Bones are united by cartilage.
   - Types:
     i. Primary Cartilaginous Joint (Synchondrosis) → Temporary, replaced by bone after growth (e.g., epiphyseal plate).
     ii. Secondary Cartilaginous Joint (Symphysis) → Permanent,fibrocartilage between bones, allows slight movement (e.g., pubic symphysis, intervertebral discs).



3. ***Synovial Joints (freely movable, most evolved)***
   - Most common and mobile type.
   - Types:
     i. Ball and Socket → Hip, shoulder.
     ii. Hinge → Elbow, ankle, knee.
     iii. Pivot → Atlanto-axial, radioulnar.
     iv. Condylar (bicondylar) → Knee (femur–tibia articulation).



v. Ellipsoid → Wrist joint.
vi. Plane → Intercarpal, intertarsal joints.
vii. Saddle (Sellar) → 1st carpometacarpal joint (thumb).

B. Functional Classification

1. ***Synarthroses*** – Immovable joints (e.g., sutures in skull).
2. ***Amphiarthroses*** – Slightly movable joints (e.g., pubic symphysis, intervertebral joints).
3. ***Diarthroses*** – Freely movable joints (all synovial joints).

C. Regional Classification

1. ***Skull Type –*** Immovable sutures.
2. ***Vertebral Type*** – Slightly movable (intervertebral discs).
3. ***Limb Type*** – Freely movable (synovial joints).

D. According to Number of Articulating Bones

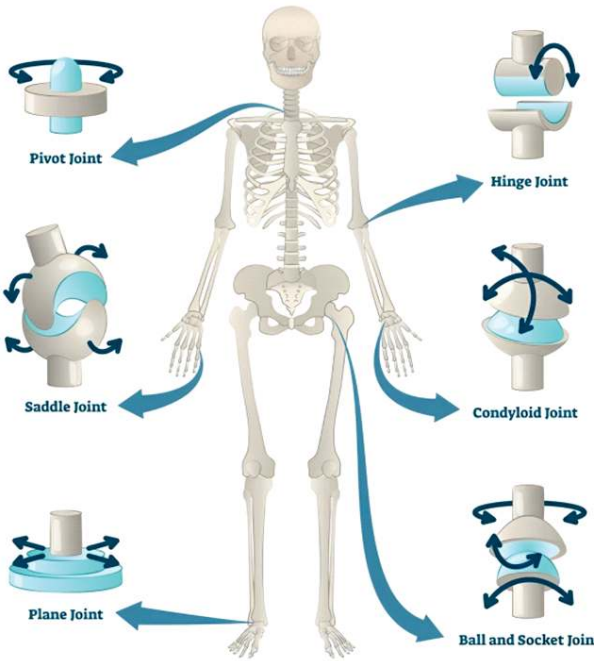
1. ***Simple Joint*** → Two bones articulate (e.g., hip, shoulder).
2. ***Compound Joint*** → More than two bones articulate in one capsule (e.g., elbow joint: humerus, radius, ulna).
3. ***Complex Joint*** → Joint cavity divided by intra-articular disc or meniscus (e.g., knee joint, temporomandibular joint).

Structure of a Synovial Joint

- Articular surfaces: Covered by hyaline cartilage (smooth, low-friction).
- Joint cavity: Contains synovial fluid.
- Synovial membrane: Produces synovial fluid, lubricates joint.
- Fibrous capsule: Encloses joint, provides strength.
- Ligaments: Strengthen and stabilize.
- Articular discs/menisci: Improve congruence and shock absorption (in some joints).
- Bursae: Fluid-filled sacs reducing friction.

Examples of Different Synovial Joints

- Ball & Socket → Shoulder, hip.
- Hinge → Elbow, ankle, interphalangeal.
- Pivot → Atlanto-axial, proximal radioulnar.
- Condylar → Knee.
- Ellipsoid → Wrist.
- Plane → Intercarpal, sternoclavicular.
- Saddle → Thumb carpometacarpal.



Functions of Joints

- Allow movements (flexion, extension, rotation, circumduction).
- Provide mechanical leverage.
- Absorb shocks during movement.
- Maintain posture and stability.

Clinical Importance

i. Dislocations → Shoulder, hip, temporomandibular joint.
ii. Arthritis → Inflammatory disease (rheumatoid, osteoarthritis).
iii. Ligament Injuries → Common in knee (ACL, PCL tears).
iv. Developmental Disorders → Congenital dislocation of hip.
v. Degenerative Changes → With age, articular cartilage thins → restricted mobility.

Synovial Joint

Q. Explain the Synovial Joint in Details

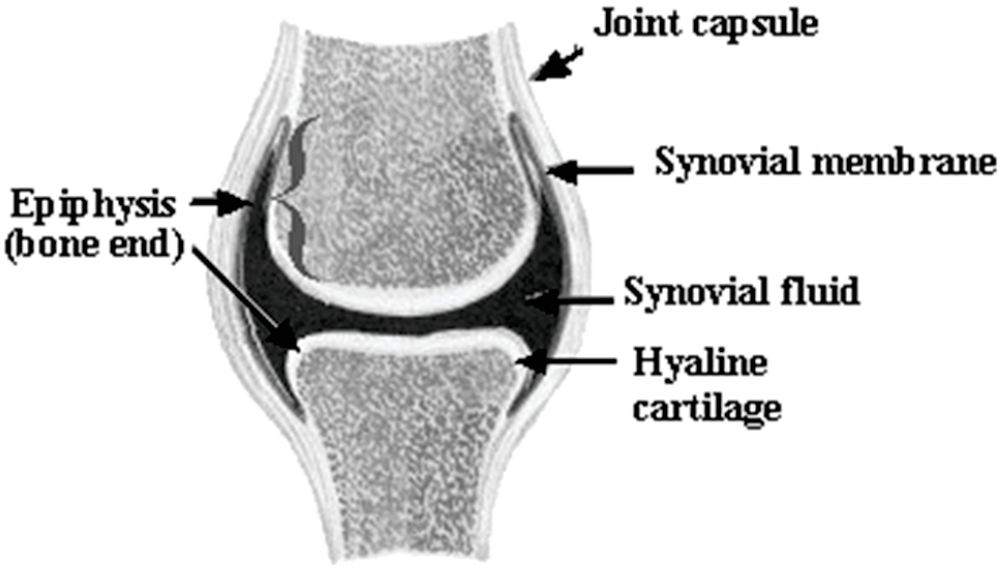
Definition

- A synovial joint is the most evolved type of joint in the human body.
- It is characterized by a joint cavity filled with synovial fluid, allowing free mobility.
- Found mostly in limbs, enabling wide range of movements.

General Features (Characters)

i. Articular Surfaces → Covered with hyaline cartilage (except TMJ, sternoclavicular joint where fibrocartilage is present).
ii. Joint Cavity → Contains small amount of synovial fluid, reducing friction.
iii. Articular Capsule → Fibrous capsule enclosing the joint, lined internally by synovial membrane.
iv. Synovial Membrane → Secretes synovial fluid (rich in hyaluronic acid), lubricating the joint.
v. Ligaments → Strengthen the capsule, maintain stability.
vi. Accessory Structures (present in some joints):
- Menisci (knee)
- Articular disc (TMJ, sternoclavicular)
- Labrum (hip, shoulder)
- Bursae and tendon sheaths (reduce friction).

Blood Supply → Rich vascular supply, forms periarticular arterial anastomosis.
Nerve Supply → From articular nerves (Hilton's law: nerve supplying a muscle crossing a joint also supplies that joint).



Classification of Synovial Joints (Based on Shape of Articular Surfaces & Movement)

1. Plane (Gliding) Joints
- Articular Surfaces: Flat or slightly curved.
- Movement: Gliding in various directions (non-axial).
- Examples: Intercarpal joints, intertarsal joints, sternoclavicular (partly), acromioclavicular joint.

2. Hinge Joints

- Articular Surfaces: Pulley-shaped, allow movement in one plane.
- Movement: Flexion and extension only (uniaxial).
- Examples: Elbow joint, ankle joint, interphalangeal joints.

3. Pivot Joints

- Articular Surfaces: A bony pivot rotates within an osteoligamentous ring.
- Movement: Rotation only (uniaxial).
- Examples: Median atlanto-axial joint, superior and inferior radioulnar joints.

4. Condylar (Bicondylar) Joints

- Articular Surfaces: Two condyles articulate with shallow cavities.
- Movement: Mainly flexion and extension, limited rotation (biaxial).
- Examples: Knee joint, temporomandibular joint (TMJ).

5. Ellipsoid Joints

- Articular Surfaces: Oval convex surface fits into oval concave surface.
- Movement: Flexion, extension, abduction, adduction, circumduction (biaxial).
- Examples: Wrist joint (radiocarpal), atlanto-occipital joint.

6. Saddle (Sellar) Joints

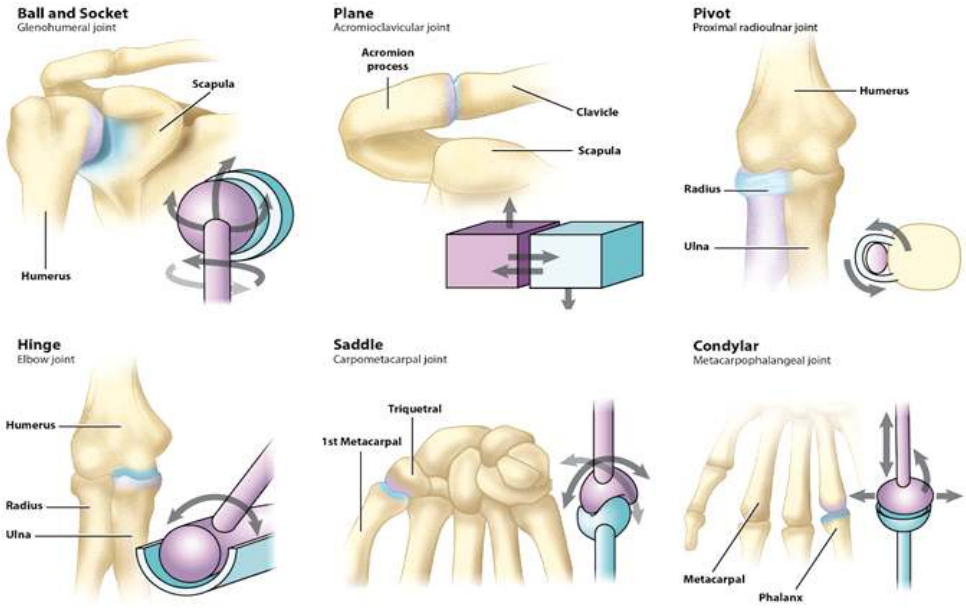
- Articular Surfaces: Reciprocal concavo-convex surfaces.
- Movement: Flexion, extension, abduction, adduction, circumduction (biaxial, but more freedom than ellipsoid).
- Examples: First carpometacarpal joint (thumb), sternoclavicular joint.

7. Ball-and-Socket Joints

- Articular Surfaces: Rounded head fits into cup-shaped socket.
- Movement: Multiaxial – flexion, extension, abduction, adduction, rotation, circumduction.
- Examples: Shoulder joint, hip joint.

Movements in Synovial Joints

- Uniaxial: Movement in one axis (hinge, pivot).
- Biaxial: Movement in two axes (ellipsoid, saddle, condylar).
- Multiaxial: Movement in multiple axes (ball and socket, plane).



Blood Supply & Nerve Supply

- Arterial: Periarticular arterial anastomosis around joint.
- Venous: Follow arteries → drain into local veins.
- Nerve Supply:
  - Hilton's Law → Nerves supplying muscles acting on a joint also supply the joint.
  - Contain sensory endings for proprioception and pain.

Clinical Importance

1. ***Arthritis***
   - Osteoarthritis → Degenerative (common in knee, hip).
   - Rheumatoid arthritis → Autoimmune.
   - Gouty arthritis → Uric acid crystal deposition.
2. ***Dislocations & Subluxations***
   - Common in shoulder and TMJ due to weak capsules.
3. ***Ligament Injuries***
   - Cruciate ligament injuries in knee.
   - Ankle sprains (lateral ligaments).
4. ***Joint Effusion***
   - Accumulation of excess synovial fluid due to infection or trauma.
5. ***Developmental Disorders***
   - Congenital dislocation of hip.

Student Key Note Space

OLD VERSION

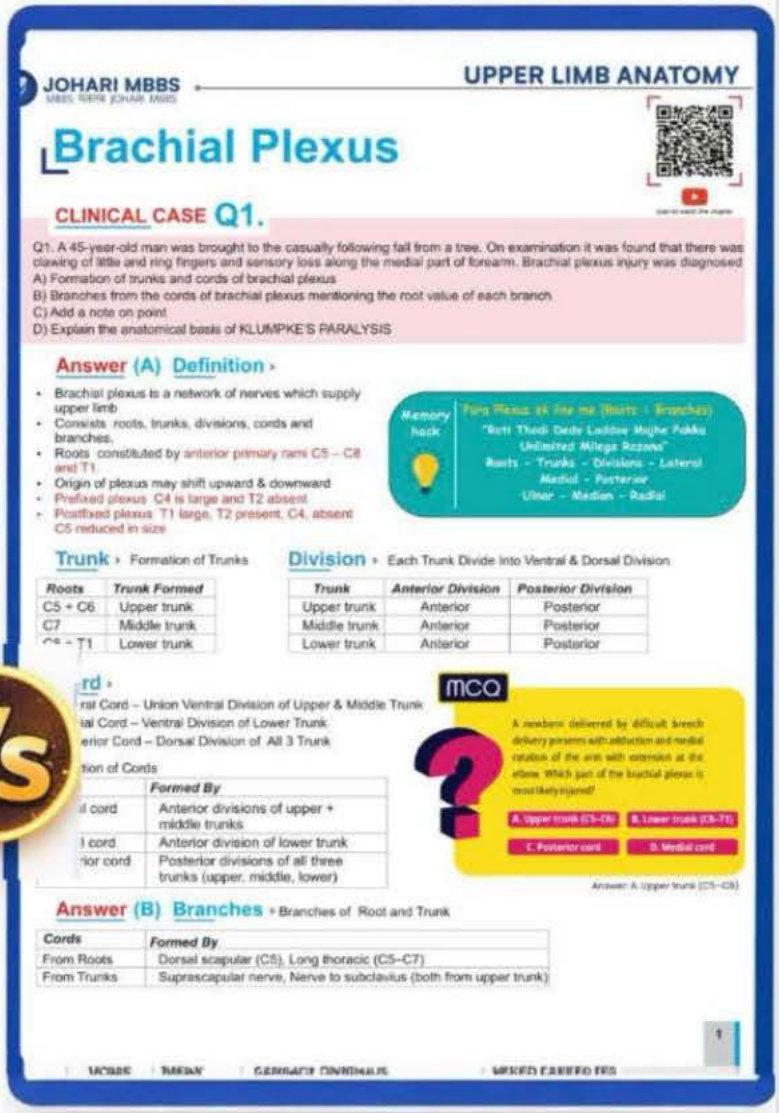
❌ No MCQ Box

VS

NEW VERSION ✅

✅ MCQ Integrated





VS

❌ No MCQ Box

❌ No Clinical Integration

❌ Plain Theory

➡

✅ MCQ Integrated

✅ Clinical Case Added

✅ Exam-Oriented Question Format

CALL : 9827765060 For More Information

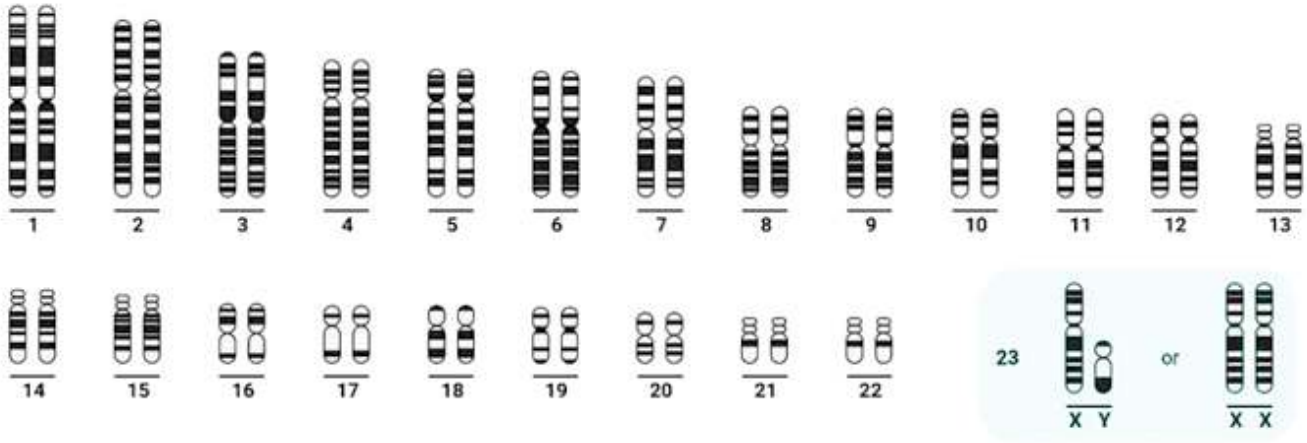
Genetics General Anatomy

Karyotyping

Q. Explain the Principle of Karyotyping

Definition

- Karyotyping is the process of arranging and analyzing chromosomes in a standard format (in pairs, decreasing order of size) from a metaphase spread of a cell.
- The standard diagrammatic representation of a set of chromosomes is called an Idiogram.



Principle

- Chromosomes are best visualized during metaphase of mitosis when they are maximally condensed.
- They are identified on the basis of:
  - *i. Length of chromosome/arms.*
  - *ii. Position of centromere.*
  - *iii. Presence of secondary constrictions/satellites.*
  - *iv. Banding pattern after staining.*

Procedure

i. Sample Collection → Usually peripheral blood lymphocytes (can also use bone marrow, amniotic fluid, chorionic villi).

ii. Culture → Lymphocytes cultured in presence of phytohaemagglutinin (PHA) to stimulate mitosis.

iii. Arrest of Cell Division → Colchicine is added to arrest cells at metaphase.

iv. Hypotonic Treatment → Causes swelling of cells, spreading chromosomes apart.

v. Fixation → Treated with fixative (methanol + acetic acid).

vi. Slide Preparation → Drops of suspension placed on slide, air-dried.

vii. Staining → Special stains (Giemsa, Quinacrine) used.

viii. Microscopy → Chromosomes spread, counted, photographed.

ix. Arrangement → Chromosomes are cut and arranged in pairs from 1–22 (autosomes) + sex chromosomes, in decreasing order of size.

Banding Techniques

- G-banding (Giemsa stain after trypsin digestion) → Produces light & dark bands, most widely used.
- Q-banding (Quinacrine stain, fluorescence microscopy).
- R-banding (Reverse banding, useful for distal chromosome ends).
- C-banding (Stains centromeric regions).
- High-resolution banding → Detects small structural abnormalities.

Applications of Karyotyping

1. ***Detection of Numerical Abnormalities***
   - Trisomy 21 (Down syndrome).
   - Monosomy X (Turner syndrome).
   - Trisomy 18 (Edward syndrome), Trisomy 13 (Patau syndrome).
   - Klinefelter syndrome (47, XXY).
2. ***Detection of Structural Abnormalities***
   - Deletions, duplications, inversions, translocations (e.g., Philadelphia chromosome in CML).
3. ***Sex Determination***
   - XX → female, XY → male.
4. ***Prenatal Diagnosis***
   - Amniocentesis, chorionic villus sampling used to detect genetic disorders.
5. ***Cancer Cytogenetics***
   - Certain cancers show specific chromosomal changes (e.g., Burkitt's lymphoma, CML).

Limitations

- Requires dividing cells (mitotic metaphase).
- Resolution is limited → cannot detect very small mutations (point mutations, microdeletions).
- Time-consuming and labor-intensive.

Clinical Importance

- Used in genetic counseling for families with inherited disorders.
- Helps in infertility workup and recurrent pregnancy loss.
- Important for diagnosis and prognosis of hematological malignancies.
- Plays role in forensic medicine for sex identification.

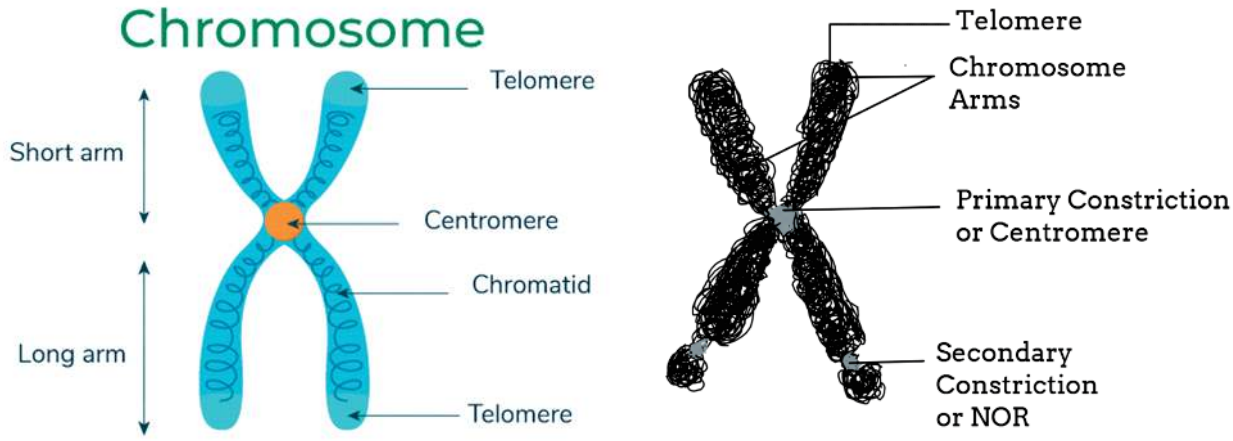
Student Key Note Space

Chromosomes

Q. Explain the Structural Anomalies of Chromosomes

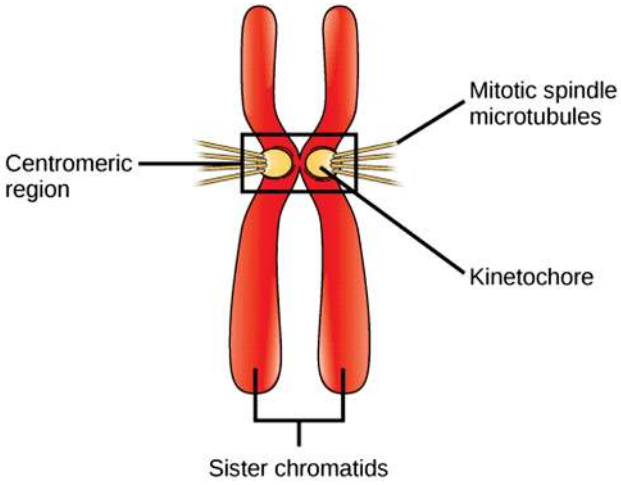
Definition

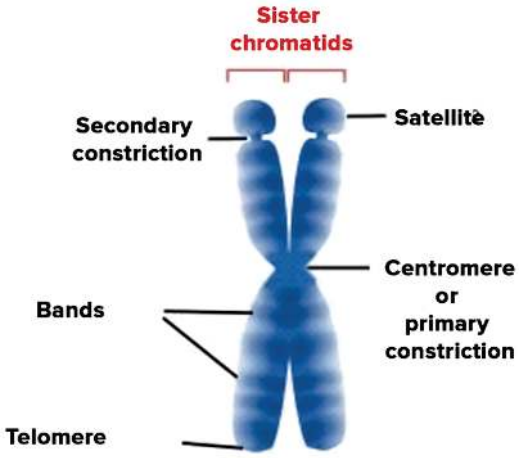
- A chromosome is a thread-like structure found in the nucleus of eukaryotic cells, composed of DNA and proteins (mainly histones).
- It is the physical carrier of genes and hence the unit of heredity.
- In humans: 46 chromosomes (23 pairs) are present → 22 pairs of autosomes + 1 pair of sex chromosomes (XX in females, XY in males).



General Anatomy of a Chromosome

1. ***Chromatid***
   - Each metaphase chromosome has two sister chromatids, joined at the centromere.
   - Each chromatid contains a double-stranded DNA molecule tightly coiled around histone proteins.
2. ***Centromere (Primary Constriction)***
   - Non-staining constricted region dividing the chromosome into two arms.
   - Essential for proper segregation of chromosomes during mitosis and meiosis.
3. ***Kinetochore***
   - A protein structure assembled at the centromere.
   - Functions as the site of spindle fiber attachment during cell division.
4. ***Arms***
   - Short arm: p arm (from French petit).
   - Long arm: q arm.
   - Arm ratio (length of q/p) is used in chromosome classification.
5. ***Telomeres***
   - DNA-protein complexes at the ends of chromosomes.
   - Protect chromosome ends from degradation and prevent fusion with other chromosomes.
   - Important in aging and cancer biology.
6. ***Secondary Constriction***
   - Constrictions other than the centromere.
   - Nucleolar Organizer Region (NOR) is a type of secondary constriction involved in nucleolus formation.
7. ***Satellite***
   - A small chromosomal fragment separated by a secondary constriction.
   - Found in acrocentric chromosomes (13, 14, 15, 21, 22).

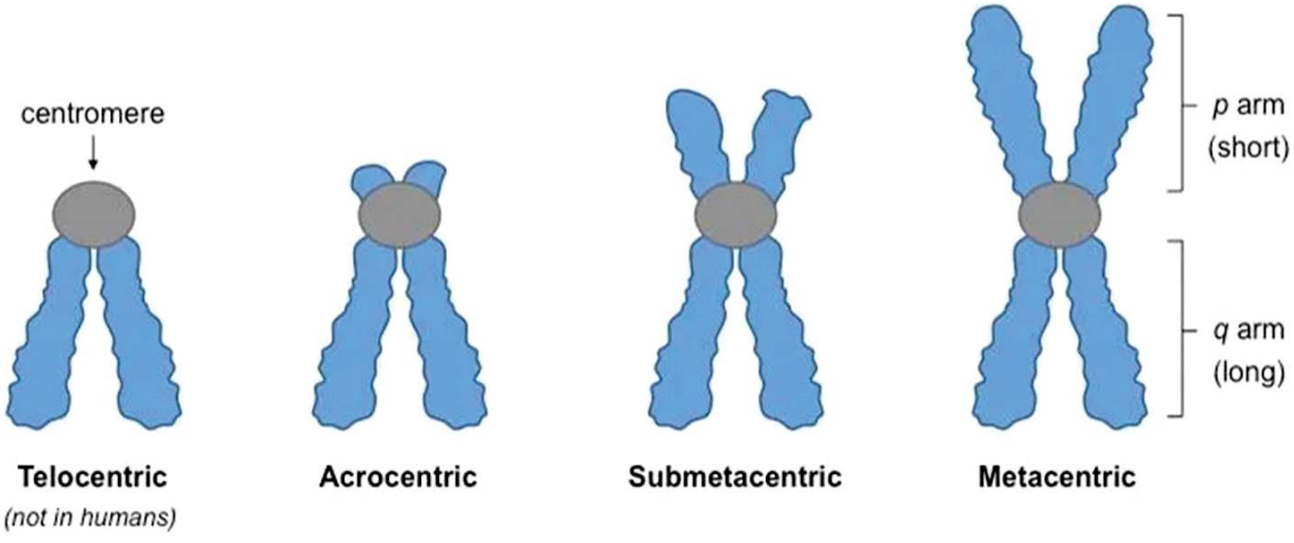




Classification of Chromosomes

A. Based on Position of Centromere

1. ***Metacentric***
   - Centromere in the middle, arms equal.
   - Example: Chromosome 1, 3.
2. ***Submetacentric***
   - Centromere slightly off-center, arms unequal.
   - Example: Chromosome 4–12.
3. ***Acrocentric***
   - Centromere near one end → one very short arm (satellite-bearing) and one long arm.
   - Example: Chromosomes 13, 14, 15, 21, 22.
4. ***Telocentric***
   - Centromere terminal, only one arm visible.
   - Not found in humans but present in rodents



B. Based on Function

- Autosomes → 44 chromosomes in humans (non-sex determining).
- Sex chromosomes → XX (female), XY (male).

C. Based on Size

- Grouped into 7 groups (A–G) in human karyotype, ranging from longest (Group A) to shortest (Group G).

Structural Anomalies of Chromosomes

Structural abnormalities arise due to chromosomal breakage and faulty repair.

1. Deletion

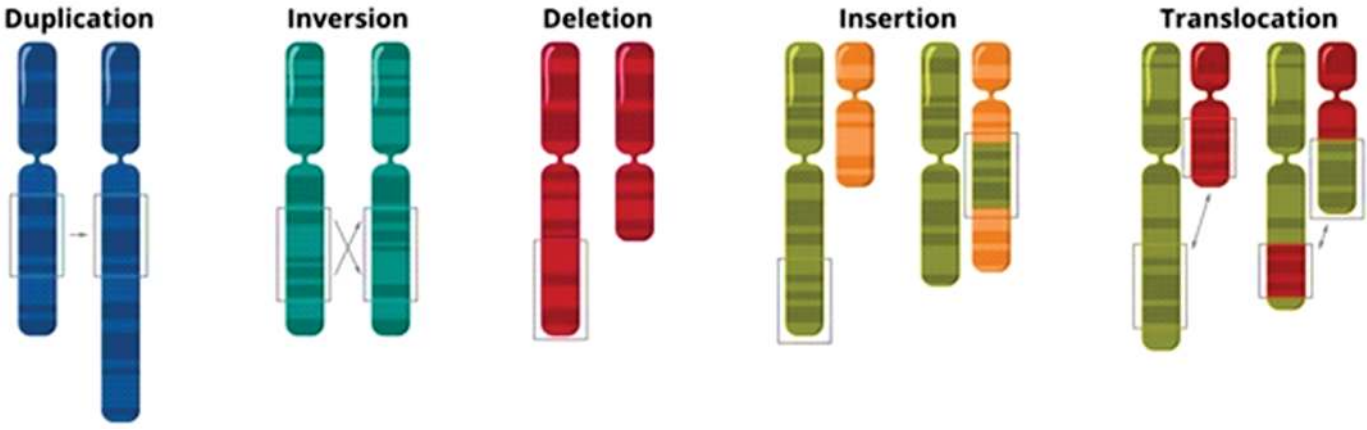
- Loss of a chromosome segment → results in partial monosomy.
- Examples:
  - Cri-du-chat syndrome (deletion of short arm of chromosome 5).
  - Wolf–Hirschhorn syndrome (deletion of short arm of chromosome 4).

2. Duplication

- Extra copy of a chromosome segment → results in partial trisomy.
- May cause gene overexpression.
- Example: Charcot–Marie–Tooth disease (PMP22 gene duplication on chromosome 17).

3. Inversion

- A chromosomal segment breaks and reinserts in reverse orientation.
- Types:
  - Paracentric: Does not involve centromere.
  - Pericentric: Involves centromere.
- May cause infertility, miscarriage due to abnormal gametes



Student Key Note Space

Sex Chromosomes

Q. Explain Structure, General Feature and disorder of sex chromosomes

Definition

- Sex chromosomes are the pair of chromosomes that determine the genetic sex of an individual.
- In humans, they form the 23rd pair of chromosomes:
  XX → Female
  XY → Male

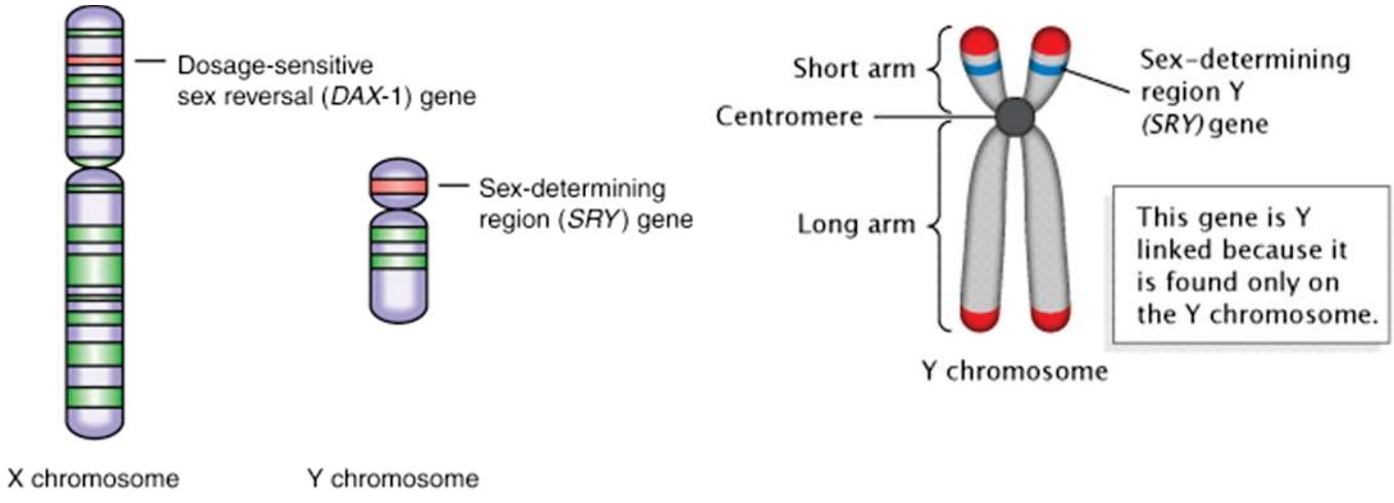
General Features

- Humans have 46 chromosomes (23 pairs) → 44 autosomes + 2 sex chromosomes.
- Sex chromosomes carry not only sex-determining genes but also many genes controlling other traits.

Structure

1. ***X Chromosome***
   - Large (about 155 million base pairs).
   - Contains ~1,100 genes.
   - Important genes:
     - XIST (X-inactivation).
     - Genes for clotting factors (Hemophilia A/B), color vision, G6PD, Duchenne muscular dystrophy.

2. ***Y Chromosome***
   - Small (about 58 million base pairs).
   - Contains ~200 genes.
   - Important genes:
     - SRY gene (Sex-determining Region Y) → directs testis formation.
     - Genes for spermatogenesis.
   - Has pseudoautosomal regions (PAR) at ends → recombine with X chromosome during meiosis.



Role in Sex Determination

- Presence of Y chromosome with SRY gene → male sex differentiation.
- Absence of Y → default development into female.
- Gonads differentiate into testis in males (testosterone & MIS secretion) and ovaries in females.

X-Inactivation (Lyon Hypothesis)

- In females (XX), one X chromosome is randomly inactivated in early embryogenesis.
- Inactivated X appears as Barr body.
- Maintains dosage compensation between males (XY) and females (XX).

Disorders of Sex Chromosomes

1. Numerical Abnormalities :

- *Turner Syndrome (45, XO) :* Female with short stature, webbed neck, streak ovaries, no Barr body.
- *Klinefelter Syndrome (47, XXY)* : Male with tall stature, gynecomastia, small testes, infertility, one Barr body.
- *Triple X Syndrome (47, XXX)* : Female, usually normal, may have learning difficulties, two Barr bodies.
- *XYY Syndrome (47, XYY) :* Male, tall, may have behavioral issues, fertile.

2. Structural Abnormalities

- Deletions or translocations in sex chromosomes can cause infertility, gonadal dysgenesis, or syndromic features.

Clinical Importance

i. Sex Determination in ambiguous genitalia.
ii. Diagnosis of Genetic Disorders (Turner, Klinefelter, etc.).
iii. Forensic Medicine → Identification of sex from cells (Buccal smear test, Barr body test).
iv. Genetic Counseling → Helps families with infertility or recurrent pregnancy loss.

Student Key Note Space

Barr Body

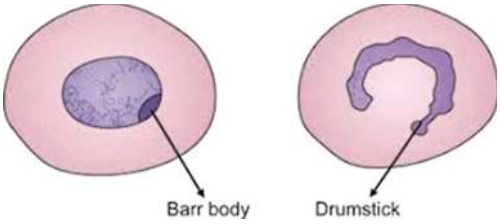
Q. What is Barr Body.

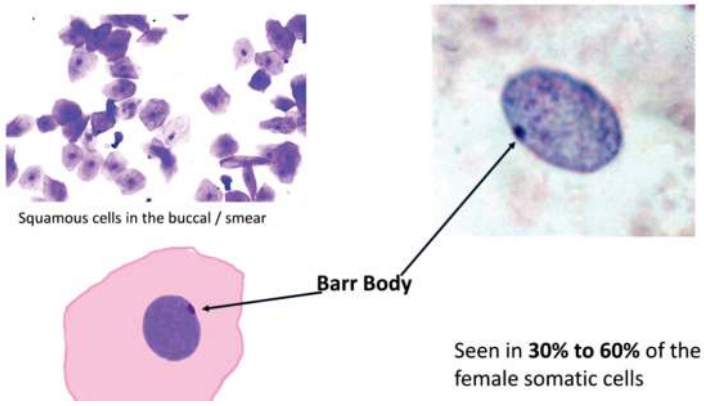
Definition

- A Barr body (also called sex chromatin) is the inactivated X chromosome present as a dense, dark-staining mass of chromatin at the periphery of the nucleus in female somatic cells.
- Discovered by Murray Barr in 1949.

Structure & Location

- Appears as a drumstick-shaped appendage in the nucleus of neutrophils or a dense spot at the nuclear membrane of epithelial cells.
- Found in:
  - i. Buccal smear cells.
  - ii. Hair root cells.
  - iii. Neutrophils (drumstick appearance).
  - iv. Amniotic fluid cells.





Formation

- In females (XX), one X chromosome is randomly inactivated early in embryogenesis (Lyon's Hypothesis of X-inactivation).
- This prevents overexpression of X-linked genes.
- Number of Barr bodies = Total X chromosomes – 1.

Normal Pattern

- Female (46, XX) → 1 Barr body.
- Male (46, XY) → No Barr body.
- Superfemale (47, XXX) → 2 Barr bodies.
- Klinefelter syndrome (47, XXY male) → 1 Barr body.
- Turner syndrome (45, XO female) → No Barr body.

Significance

1. ***Sex Determination :*** Useful in identifying genetic sex in doubtful cases (e.g., sports, forensic medicine).
2. ***Diagnosis of Chromosomal Disorders :*** Presence/absence helps in identifying sex chromosome anomalies:
   - Klinefelter (XXY) → Barr body present.
   - Turner (XO) → Barr body absent.
3. ***Genetic Counseling :*** Assists in counseling families with chromosomal syndromes.
4. ***Forensic Applications :*** Buccal smear test for Barr body helps in sex identification in legal cases.

Limitations

- Not 100% reliable: some mosaic cases may give misleading results.
- More accurate tests (Karyotyping, FISH, PCR) are now preferred.

Mendelian Inheritance

Q. Explain in Detail Mendelian Inheritance

Definition

- Mendelian inheritance refers to the patterns of transmission of traits from parents to offspring as first described by Gregor Mendel in 1865, based on his pea plant experiments.
- It is governed by genes, which exist in pairs of alleles, and segregate in predictable ways during gamete formation.

Mendel's Laws

1. Law of Dominance

- In a heterozygote, one allele (dominant) expresses itself, while the other (recessive) is masked.
- Example: Tall (T) is dominant over dwarf (t) in pea plants.

2. Law of Segregation (Purity of Gametes)

- Allele pairs separate during gamete formation, so each gamete carries only one allele.
- Example: A Tt plant produces gametes T and t in equal proportion.

3. Law of Independent Assortment

- Alleles of different genes assort independently during gamete formation.
- Traits are inherited independently of each other (only true for genes on different chromosomes or far apart on same chromosome).
- Example: Inheritance of seed color and seed shape in pea.

Mendelian Patterns of Inheritance in Humans

1. Autosomal Dominant Inheritance

- Only one copy of mutant allele required to express trait.
- Transmitted vertically (every generation).
- Examples:
  - o Achondroplasia
  - o Marfan syndrome
  - o Huntington's disease

Features:

- Both sexes affected equally.
- 50% chance of passing to offspring.

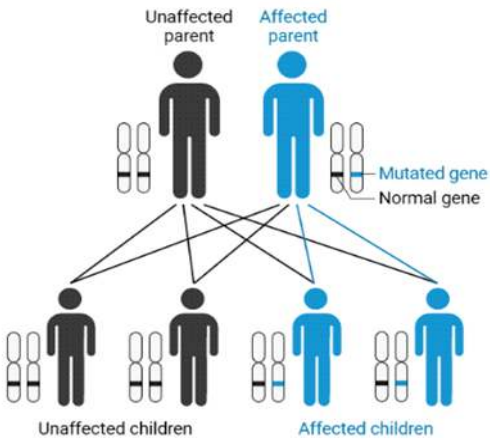
2. Autosomal Recessive Inheritance

- Two mutant alleles needed for expression.
- Carriers (heterozygotes) are unaffected.
- Examples:
  - i. Cystic fibrosis
  - ii. Phenylketonuria
  - iii. Sickle cell anemia

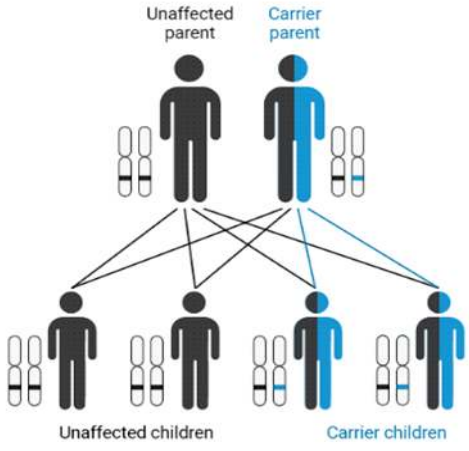
Features:

- Often seen in consanguineous marriages.
- Horizontal transmission (siblings affected).

Autosomal dominant inheritance



Autosomal recessive inheritance



3. X-linked Dominant Inheritance

- Mutant gene on X chromosome, expressed in heterozygotes.
- Examples: Fragile X syndrome, Vitamin D–resistant rickets.

Features:

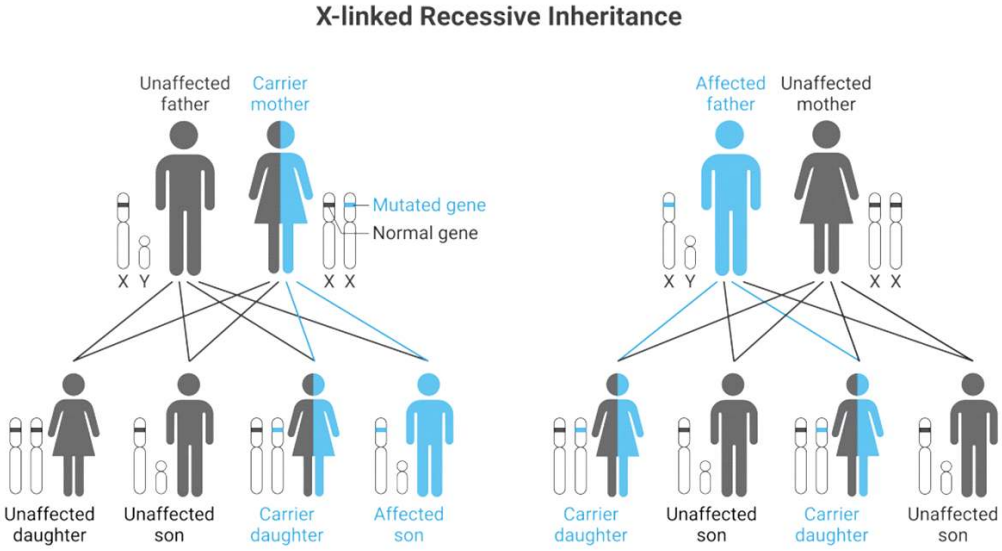
- Both sexes affected, but females more often.
- Affected male transmits to all daughters, no sons.

4. X-linked Recessive Inheritance

- Mutant gene on X chromosome, expressed in males (since XY).
- Examples: Hemophilia A & B, Duchenne muscular dystrophy, G6PD deficiency.

Features:

- Males predominantly affected.
- Carrier mothers → 50% chance of affected sons



5. Y-linked (Holandric) Inheritance

- Gene carried only on Y chromosome.
- Examples: Hairy pinna, some forms of male infertility.

Features:

- Transmitted from father to all sons.
- Females never affected.

6. Mitochondrial Inheritance (Non-Mendelian, Maternal)

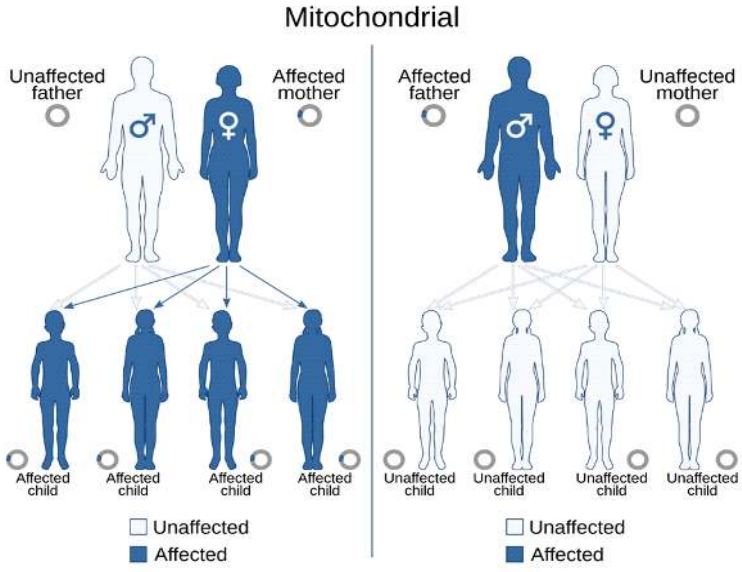
- Genes in mitochondria are inherited only from mother.
- Examples: Leber's hereditary optic neuropathy.
- Both sexes affected, but only mothers transmit.

Exceptions to Mendelian Inheritance

- Incomplete dominance → Blended phenotype (e.g., pink flower from red × white).
- Codominance → Both alleles expressed (e.g., ABO blood group).
- Multiple alleles → More than 2 alleles exist (e.g., ABO system).
- Polygenic inheritance → Trait controlled by multiple genes (e.g., height, skin color).
- Pleiotropy → One gene influences multiple traits (e.g., Marfan syndrome).
- Epistasis → One gene masks effect of another gene.

Clinical Significance

- Helps in predicting recurrence risk in families.
- Useful in genetic counseling for hereditary diseases.
- Explains inheritance of metabolic disorders, hemoglobinopathies, and skeletal anomalies.
- Forms basis for pedigree analysis in medicine.



Numerical Aberrations of Chromosomes

Q. Write a short note on Numerical Aberrations of Chromosomes

Definition

- Numerical aberrations are abnormalities in the number of chromosomes due to errors in cell division (non-disjunction, anaphase lag, polyploidy).
- They result in loss or gain of chromosomes → leading to genetic syndromes.

Types

1. Aneuploidy

- Gain or loss of one or more individual chromosomes (not a whole set).
- Usually arises from non-disjunction during meiosis.

a) Monosomy (2n – 1)

- Loss of a single chromosome.
- Example: Turner syndrome (45, XO).

b) Trisomy (2n + 1)

- Presence of an extra chromosome.
- Examples:
  - Down syndrome (Trisomy 21).
  - Edwards syndrome (Trisomy 18).
  - Patau syndrome (Trisomy 13).

c) Tetrasomy (2n + 2)

- Extra copy of a chromosome pair (rare).

d) Nullisomy (2n – 2)

- Loss of a homologous chromosome pair → usually lethal in humans.

2. Polyploidy

- Abnormal increase in the number of complete sets of chromosomes.
- Normal human diploid = 2n = 46.
- Polyploidy = triploidy (3n = 69) or tetraploidy (4n = 92).
- Usually lethal in humans → incompatible with life.

Causes

i. Non-disjunction → failure of homologous chromosomes or chromatids to separate during meiosis/mitosis.

ii. Anaphase lag → delayed movement of chromosome/chromatid, excluded from nucleus.

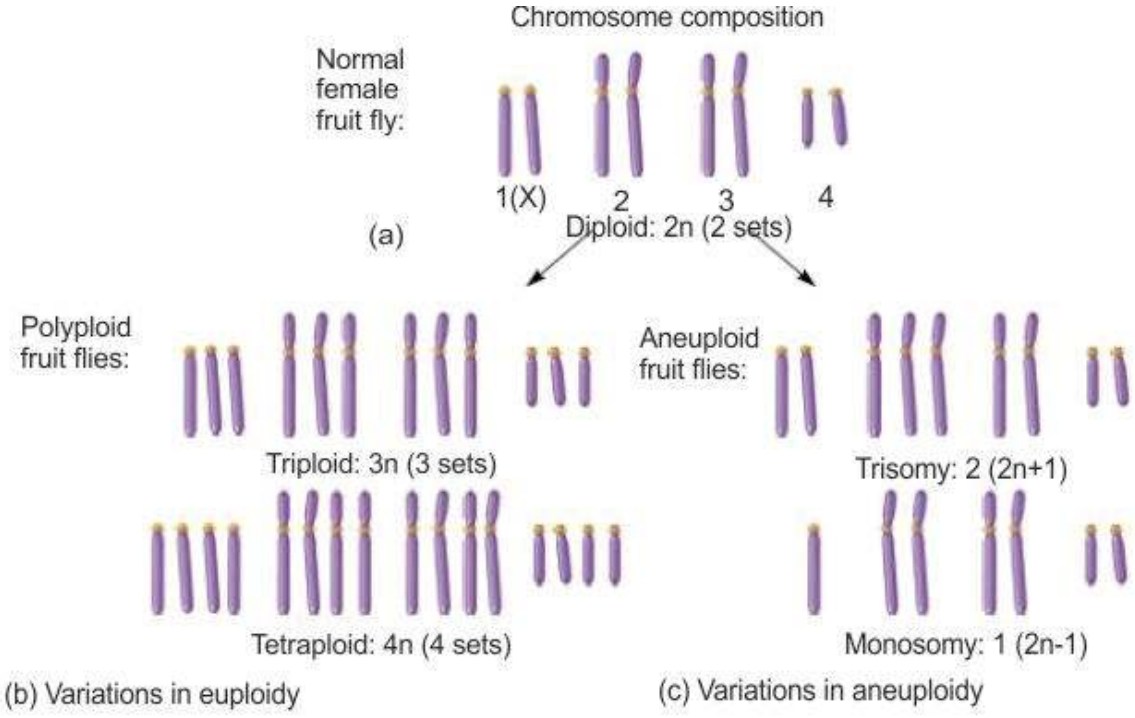
iii. Fertilization errors → polyspermy, diploid gametes.

Examples of Human Disorders

Condition	Chromosomal Change	Features
Down Syndrome	Trisomy 21	Mental retardation, epicanthic folds, single palmar crease, congenital heart disease.
Edward Syndrome	Trisomy 18	Severe intellectual disability, clenched fists, rocker-bottom feet.
Patau Syndrome	Trisomy 13	Cleft lip/palate, microphthalmia, polydactyly, heart defects.
Turner Syndrome	Monosomy X (45, XO)	Short stature, webbed neck, primary amenorrhea, infertility.
Klinefelter Syndrome	47, XXY	Tall stature, gynecomastia, small testes, infertility.
Triploidy	69 chromosomes (3n)	Spontaneous abortion, rarely live-born.
Tetraploidy	92 chromosomes (4n)	Lethal, early embryonic death.

Clinical Importance

- Major cause of congenital anomalies and mental retardation.
- Leading cause of spontaneous abortions (about 50% of miscarriages due to chromosomal abnormalities).
- Important in prenatal diagnosis (amniocentesis, chorionic villus sampling).
- Helps in genetic counseling for affected families.



Types of variation in chromosome number. (a) Depicts the normal diploid number of chromosomes in Drosophila. (b) Examples of polyploidy. (c) Examples of aneuploidy.

Variation in chromosomes number

Chromosomal Disorders

1. Down's Syndrome (Trisomy 21)

Definition

- Most common autosomal chromosomal abnormality.
- First described by Langdon Down (1866).

Genetics

- Trisomy of chromosome 21 → total 47 chromosomes (47,XX,+21 or 47,XY,+21).
- Causes:
  i. 95% → meiotic nondisjunction (maternal meiosis I error).
  ii. 4% → Robertsonian translocation (extra chromosome 21 attached to 14 or 22).
  iii. 1% → Mosaicism (some normal cells, some trisomy).

Risk Factors

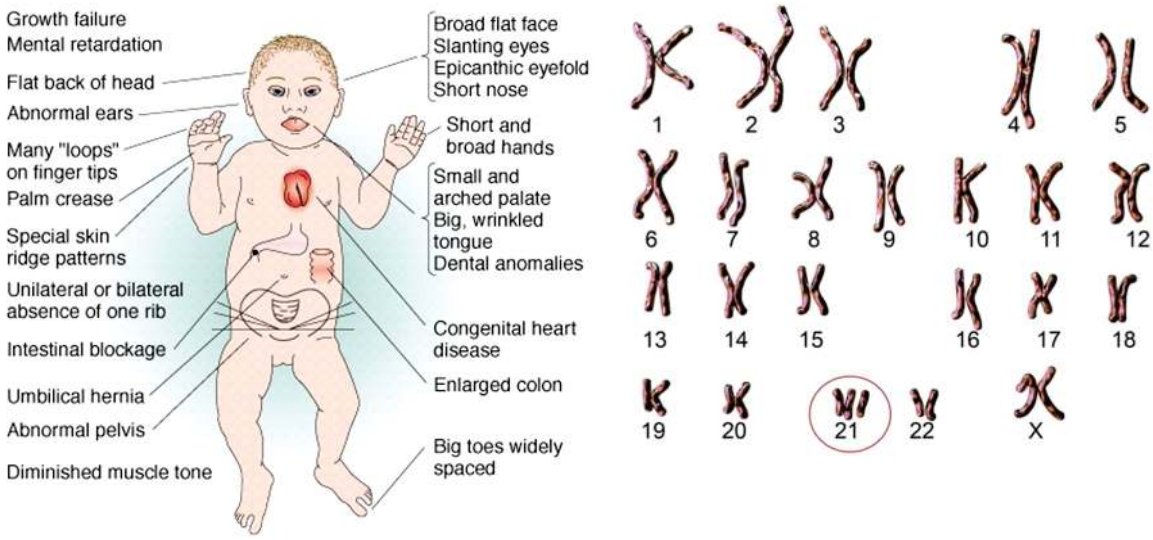
- Advanced maternal age (>35 years).
- History of chromosomal abnormality in previous child.

Clinical Features

- ***General Features:***
  i. Mental retardation / intellectual disability.
  ii. Poor growth, short stature.
  iii. Hypotonia (floppy baby).
  iv. Widely spaced great toe.
- ***Facial Features:***
  i. Flat nasal bridge, epicanthic folds.
  ii. Upward slanting palpebral fissures.
  iii. Moon face, protruding tongue, small ears.
- ***Hand Features:***
  i. Single transverse palmar crease (simian crease).
  ii. Short, broad hands.
- ***Other Features:***
  i. Congenital heart disease (endocardial cushion defects).
  ii. Increased risk of leukemia (ALL, AML).
  iii. Early-onset Alzheimer's disease.

Prenatal Diagnosis

- Screening: Increased inhibin A, increased β-hCG, decreased AFP.
- Confirmatory: Amniocentesis or chorionic villus sampling + karyotyping.



2. Klinefelter's Syndrome (47,XXY)

Definition

- Sex chromosome abnormality with one or more extra X chromosomes in a male.

Genetics

- Karyotype: 47,XXY (classic).
- Cause: Meiotic nondisjunction (maternal or paternal).

Clinical Features

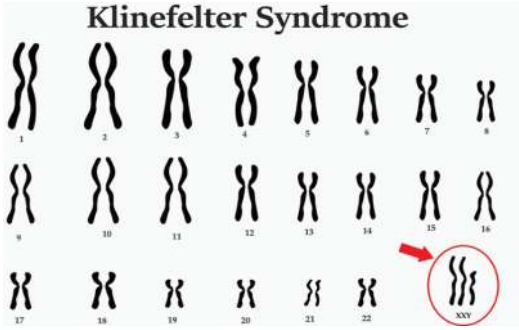
- Small testes → infertility (azoospermia).
- Hypogonadism → low testosterone, high FSH/LH.
- Gynecomastia.
- Tall stature, long limbs (eunuchoid habitus).
- Sparse pubic and facial hair.
- Mild intellectual disability or learning difficulty.
- Barr body present.

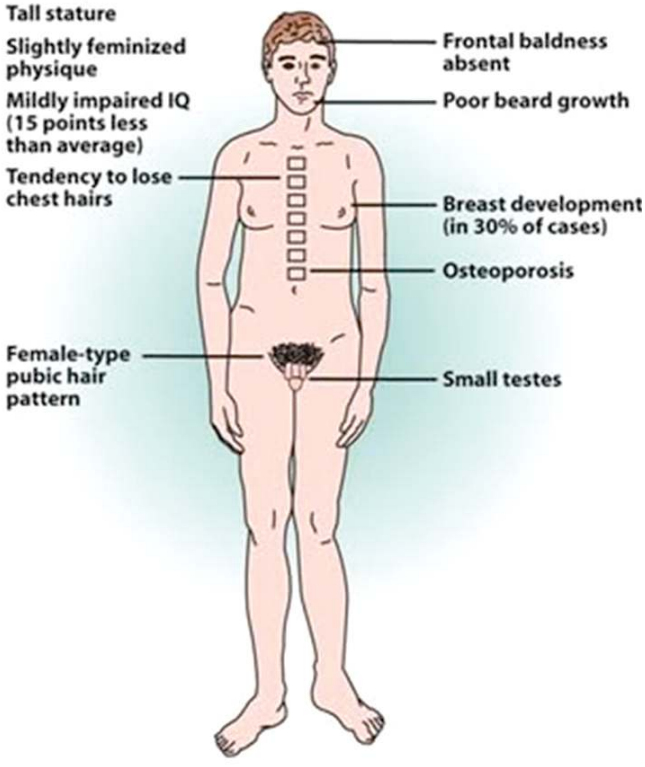
Complications

- Increased risk of breast cancer and osteoporosis.

Treatment

- Testosterone replacement therapy.
- Assisted reproduction (IVF with sperm retrieval).





Klinefelter syndrome (XXY

3. Turner's Syndrome (45,XO)

Definition

- Sex chromosome abnormality due to partial or complete loss of one X chromosome in females.

Genetics

- Karyotype: 45,XO.
- May also occur due to mosaicism (45,X/46,XX).

Clinical Features

- Short stature.
- Webbed neck, low posterior hairline.
- Shield chest with widely spaced nipples.
- Cubitus valgus (increased carrying angle).
- Primary amenorrhea (due to streak ovaries).
- Rudimentary ovaries → infertility.
- Normal intelligence (may have learning difficulty).
- No Barr body present.

Complications

- Cardiovascular anomalies (coarctation of aorta, bicuspid aortic valve).
- Renal anomalies (horseshoe kidney).
- Increased risk of autoimmune thyroiditis.

Treatment

- Growth hormone therapy (for short stature).
- Estrogen replacement therapy (for secondary sexual characteristics).
- IVF for fertility in selected cases.

Comparison Table

Disorder	*Karyotype*	*Sex*	*Key Features*
Down's	47, +21	M/F	Intellectual disability, flat facies, simian crease, CHD
Klinefelter's	47, XXY	Male	Small testes, gynecomastia, infertility, tall stature
Turner's	45, XO	Female	Short stature, webbed neck, streak ovaries, amenorrhea

